Universal Domain Adaptation Meets Multimodality Remote Sensing Data for Scene Classification

Yanbing Shen, Jianmeng Li, *Graduate Student Member, IEEE*, Zengmao Wang, *Member, IEEE*, Shiying Sun, and Bo Du, *Senior Member, IEEE*

***Abstract*—Universal domain adaptation (DA) is a practical technique for remote sensing scene classification, since the expensive collection of labels in remote sensing scenarios usually results in the relationship of the label sets between the source domain and the unlabeled target domain being unknown. The domain shift and the unknown category shift are two important challenges for the universal DA (UniDA) task. To reduce these shifts, we propose a multimodal UniDA (MUniDA) method for remote sensing scene classification by using both the source images and their captions. MUniDA first aligns the source images and the textual captions with CLIP model to mitigate the gap of cross modalities. Two classifiers are trained following the visual encoder and text encoder of CLIP with source images and their captions, respectively. Thus, the representation of the image can be much more discriminative for classification due to the rich information in the textual captions. Then a target fine-tuning method is developed with an orthogonal feature decomposition strategy and a pseudolabeling strategy. The decomposition strategy is designed by decomposing the class prototype of source images with singular vector decomposition into the orthogonal base vectors of the known class and the unknown class, which are useful to distinguish the common data into target samples into common data and private data. Meanwhile, the pseudolabeling strategy is proposed by assigning confident pseudolabels to the common data with common labels while assigning an unknown label to the private data with different classifier predictions for fine-tuning. Experimental results demonstrate that our proposed method outperforms the state-of-the-art UniDA methods.**

Received 6 June 2025; revised 23 November 2025; accepted 6 January 2026. Date of publication 12 January 2026; date of current version 27 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62271357, Grant 62476204, and Grant 62225113; in part by the Innovative Research Group Project of Hubei Province under Grant 2024AFA017; in part by the National Science and Technology Major Project under Grant 2024ZD1600100; in part by Wuhan Natural Science Foundation under Grant 2024040801020236; and in part by the Supercomputing Center of Wuhan University through Supercomputing System. *(Yanbing Shen and Jianmeng Li contributed equally to this work.) (Corresponding authors: Zengmao Wang; Shiying Sun.)*

Yanbing Shen and Jianmeng Li are with the School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: 2020302111025@whu.edu.cn; 2024102110088@whu.edu.cn).

Zengmao Wang and Bo Du are with the National Engineering Research Center for Multimedia Software, School of Computer Science, Institute of Artificial Intelligence, and Hubei Key Laboratory of Multimedia and Network Communication Engineering, Wuhan University, Wuhan 430072, China (e-mail: wangzengmao@whu.edu.cn; dubo@whu.edu.cn).

Shiying Sun is with the State Key Laboratory of Management and Control for Complex Systems, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China (e-mail: sunshiying2013@ia.ac.cn).

Digital Object Identifier 10.1109/TGRS.2026.3653436

***Index Terms*—Feature decomposition, multimodality, remote sensing, scene classification, universal domain adaptation (DA).**

I. Introduction

REMOTE sensing image scene classification is an important task for the interpretation of geospatial data to effectively understand the semantic content [1]. Deep learning has significantly advanced remote sensing scene classification, and many methods have achieved satisfactory performance [2], [3], [4]. Generally, these methods usually depend on the large amount of labeled data. However, annotating large-scale remote sensing images is expensive, primarily due to the extensive human involvement required for accurate labeling and data investigation. Domain adaptation (DA) serves as a fundamental approach to tackle this issue by leveraging knowledge from a labeled source domain to facilitate learning in a different yet related unlabeled target domain [5].

With the relationship of label spaces between the source domain and target domain, the existing DA approaches can be divided into five categories, i.e., closed-set DA, partial DA, open-set DA, open-partial DA, and universal DA (UniDA) [6], [7], [8], [9], [10], [11]. For convenience, we define the label set of the source domain as Y_s and that of the target domain as Y_t. Closed-set DA assumes a strict constraint where the source and target domains share the same label set ($Y_s = Y_t$). Partial DA and open-set DA relax this constraint. In partial DA, the target label space is a subset of the source label space ($Y_t \subseteq Y_s$), while in open-set DA, the source label space is a subset of the target label space ($Y_s \subseteq Y_t$). Open-partial DA arises when the label spaces partially overlap ($Y_s \cap Y_t \neq \emptyset$) but also have unique labels ($Y_s - Y_t \neq \emptyset$ and $Y_t - Y_s \neq \emptyset$). While for UniDA, it imposes no constraints on the relationship between the label spaces, making it the most practical approach for real-world scenarios since the label space of the target domain is often unknown. Meanwhile, since it is difficult to transfer knowledge from the source domain to the target domain, especially in scenarios where the label space of the target domain is entirely unknown. UniDA is also widely regarded as the most challenging task of DA due to domain shift and unknown category shift. In this article, we concentrate on addressing the challenges of UniDA for remote sensing scene classification.

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

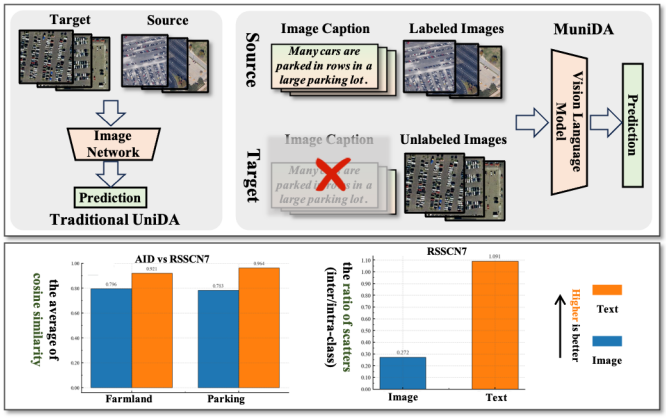


Fig. 1. First row is the paradigm of the traditional UniDA and the proposed MUniDA. The second row shows the cross domain similarities of the scenes Farmland and Parking in AID and RSSCN7 with images and textual data, respectively (left) and the relationship of interclass and intraclass scatters for the images and textual data, respectively (right).

Most of the existing UniDA methods for remote sensing scene classification primarily focus on the images [11], [12], [13], and they always ignore the ability of textual data, i.e., the captions of the source images. We found that the captions are useful to reduce the domain shift and helpful to capture the unknown category shift. We show a simple case study in Fig. 1. It shows the cosine similarities cross the domains of the same scene with images and textual data on AID (Farmland and parking) and RSSCN7 (field and parking), which can represent the domain gap at the image level and text level effectively. We can observe that the domain gap between images of the same scene from different domains is larger than that of textual data. This demonstrates that the textual data can bridge the domain gaps more effectively. Meanwhile, in Fig. 1, we also show the ratio between interclass scatter and intraclass scatter with images and textual data on RSSCN7, respectively. If the interclass scatter is large, it indicates that the classes are easy to distinguish. While if the interclass scatter is small, it indicates the samples of the same scene are very similar. To guarantee that the samples of different classes are easy to distinguish, we hope the interclass scatter should be large while the intraclass should be small. Hence, if the ratio between interclass scatter and intraclass scatter is large, it will be easy to deal with the category shift problem. We can observe that the ratio of textual data is larger than that of the images, which indicates that the textual data have advantages in dealing with the category shift problem. Therefore, in this article, we naturally consider introducing the captions of source images for remote sensing scene classification in UniDA scenario. Meanwhile, vision-language models have experienced rapid development and promising ability in recent years, which has provided a good chance for UniDA with both images and textual data, such as CLIP [14].

In this article, we propose a novel multimodal UniDA (MUniDA) with remote sensing images and their captions for scene classification, termed as MUniDA. In MUniDA, it mainly contains two steps. The first step is the alignment of images and captions, which aligns the images and captions of source domain based on the CLIP. To efficiently enhance the image representation with its caption, we design two classifiers following the visual encoder and the text encoder of CLIP. During the alignment, the CLIP is frozen, and we train the classifiers not only with cross entropy but also contrastive loss. Since there are no captions for target predictions, we regard the classifier trained with textual data as the auxiliary classifier and the other one trained with images as the primary classifier. Then we propose a target fine-tuning method to adapt the primary classifier for the target domain with pseudolabels. Specifically, for any kind of DA task, the alignment is an important strategy. Without loss of generality, we attempt to align the common classes between the source domain and the target domain, and it is necessary to distinguish the known class and unknown class in the target domain. An orthogonal feature decomposition strategy is developed by decomposing the class prototype of source images with singular vector decomposition into the orthogonal base vectors of the known class and the unknown class. We project the target samples onto orthogonal base vectors corresponding to the known class and the unknown class, respectively. The distribution discrepancy is adopted to distinguish target samples into common data and private data. Meanwhile, a pseudolabeling strategy is also developed based on the predictions of the primary classifier and the auxiliary classifier to assign confident pseudolabels to the common data with common labels while assigning unknown label to the private data. Extensive experiments show that the proposed MUniDA is promising for remote sensing scene classification. The main contributions of the article can be summarized as follows.

1) A novel MUniDA method is proposed for remote sensing scene classification based on CLIP by exploiting the semantic contents in images with textual data.
2) We develop a target fine-tuning method by designing an orthogonal feature decomposition strategy to distinguish the common data and private data, and a pseudolabeling strategy to assign confident pseudolabels to common data and the private data.
3) Extensive experiments on UniDA task with several popular and benchmark datasets show that the proposed method has achieved significantly improvement for scene classification with multimodality remote sensing data.

The rest of the article is organized as follows. Section II provides a review of the related work about DA. Section III introduces the details of the methodology. Section IV presents the results analysis with several popular and benchmark datasets. Section V concludes the article with a brief summary.

II. Related Work

A. Traditional DA

Most existing DA settings for remote sensing image scene classification can be categorized into closed-set, partial, and open-set DA based on the label-set relationship.

For closed-set DA, source domain and target domain have the same label space. Lots of works have fully explored closed-set DA. DAN [15] introduces a DA

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

network that finetunes a pretrained CNN by jointly minimizing source cross-entropy, source–target distribution discrepancy, and target geometric structure, demonstrating strong cross-sensor and cross-region classification robustness. AANN [16] is an asymmetric adaptation neural network that first compresses pretrained CNN features via a denoising autoencoder and then aligns source-to-target distributions while enforcing class separability. Recent closed-set DA methods harness the generative adversarial networks to align distributions. CaGAN [17] introduces a class-aware GAN that alternates global adversarial alignment with category-wise center matching to explicitly shrink intraclass and widen interclass distances across unlabeled multisource remote-sensing imagery. Moreover, [18] use generative adversarial networks to handle and pixel-wise classification. Recent perspectives, such as the survey [19], argue that global alignment strategies often fail in high-resolution imagery. To address this, advanced frameworks focus on disentangling domain-invariant features. For instance, DS-DWTGAN [20] integrates output adaptation and data enhancement training strategies to reinforce the acquisition of domain-invariant features, while De-GLGAN [21] employs a decomposition-based global-local transformer to align semantic segments. In the hyperspectral domain, CPDIC [22] utilizes calibrated prototypes to handle sensor variations. Furthermore, to capture topological consistency across domains, GeIraA [23] and TGP-CFSL [24] introduce graph-based embedding strategies that align scenes based on structural relationships rather than just pixel statistics.

However, it is difficult to ensure that the source domain and the target domain have the same label space. Thus, partial DA and open-set DA are proposed to relax this limitation. Partial DA handles the case where the target classes are a subset of source classes. ETN [25] introduces adversarial partial-DA by jointly learning domain-invariant features and a progressive weighting scheme that automatically up-weights shared-class source examples while down-weighting outlier classes. BA^{3}US [26] alleviates negative transfer and uncertainty propagation in partial unsupervised DA by balanced adversarial alignment that randomly augments the target with source samples to equalize label distributions and adaptive uncertainty suppression that down-weights uncertain source samples via a complement-entropy loss. Moreover, PPLM-Net [27] mitigates negative transfer through Partial Patch Local Masking, while PRR-DA [28] rectifies prototypes to filter outlier source classes. The open set DA is proposed to reduce the distribution discrepancy of the same classes in different domains and to enlarge the distribution discrepancy of different classes in different domains. MGCN [29] tackles open-set, cross-domain few-shot RS scene classification by aligning domains via feature-dispersion-weighted adaptation and refining embeddings with a multiorder graph convolutional network. ASTA [30] adaptively separates knowns from unknowns, and OSDA-ST [31] prevents overfitting to unknowns via selective thresholding.

Despite the utility of partial DA and open-set DA, these approaches—rely on prior knowledge of the category overlap, which limits their flexibility when deployment conditions are unpredictable or when the semantic boundaries of the target domain are entirely unknown.

B. Universal DA

Compared with approaches such as closet-set DA, partial DA, open-set DA, and open-partial DA, which are tailored to specific scenarios, UniDA provides a more general solution that accommodates arbitrary label space discrepancies without prior knowledge of the target domain. UniDA was first introduced to remote sensing image scene classification by Xu et al. [11]. They remove all the constraints on the relationship between label sets of the source and target domains. Subsequent works have focused on refining the "unknown" rejection mechanism. For instance, C3DA [12] propose a comprehensive C3 criterion, considering confidence, consistency and certainty of samples to better identify unknown classes in remote sensing images.

In the context of hyperspectral image (HSI) and complex scenes, UniDA methods have incorporated more sophisticated feature learning. HyUniDA [32] introduces the first UniDA framework tailored for cross-scene HSI classification. It jointly learns common-class correspondences via shared semantic pairing (SSP) and automatically infers the number of target clusters through a domain similarity score (DSS). To further enhance feature robustness, DCmix [33] presents a dual-classifier framework that synergizes closed-set and open-set classifiers to accurately label both known and unknown target classes. It introduces cross-domain feature mixup to mimic unseen categories and enforces consistency between the two classifiers. Addressing data privacy, SFMT-UDA [34] enables source-free adaptation for Local Climate Zones, a direction also explored for segmentation by SF-DANS [35].

Despite these advancements, most existing UniDA methods still rely only on visual features and overlook the rich textual information inherently available in remote sensing datasets. This omission limits their ability to fully exploit semantic context, leading to less accurate predictions, especially in complex scenes where visual features alone are insufficient for reliable classification.

III. Methodology

In this section, we will introduce the details of the proposed method and show the flowchart of the proposed method in Fig. 2. There are two steps in the proposed method. The first step is alignment of images and captions, which aligns the images and captions in source domain with CLIP. In this step, we use a contrastive loss to align the modalities. Meanwhile, we also design two classifiers following the CLIP and train them with images and textual data, respectively. As described above, we define the classifier trained with textual data as the auxiliary classifier and the classifier trained with images as the primary classifier. The second step is target fine-tuning, which adapts the primary classifier for target domain classification by fine-tuning with pseudolabels. To obtain confident pseudolabels, an orthogonal feature decomposition strategy is proposed to distinguish common data and private data, and a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

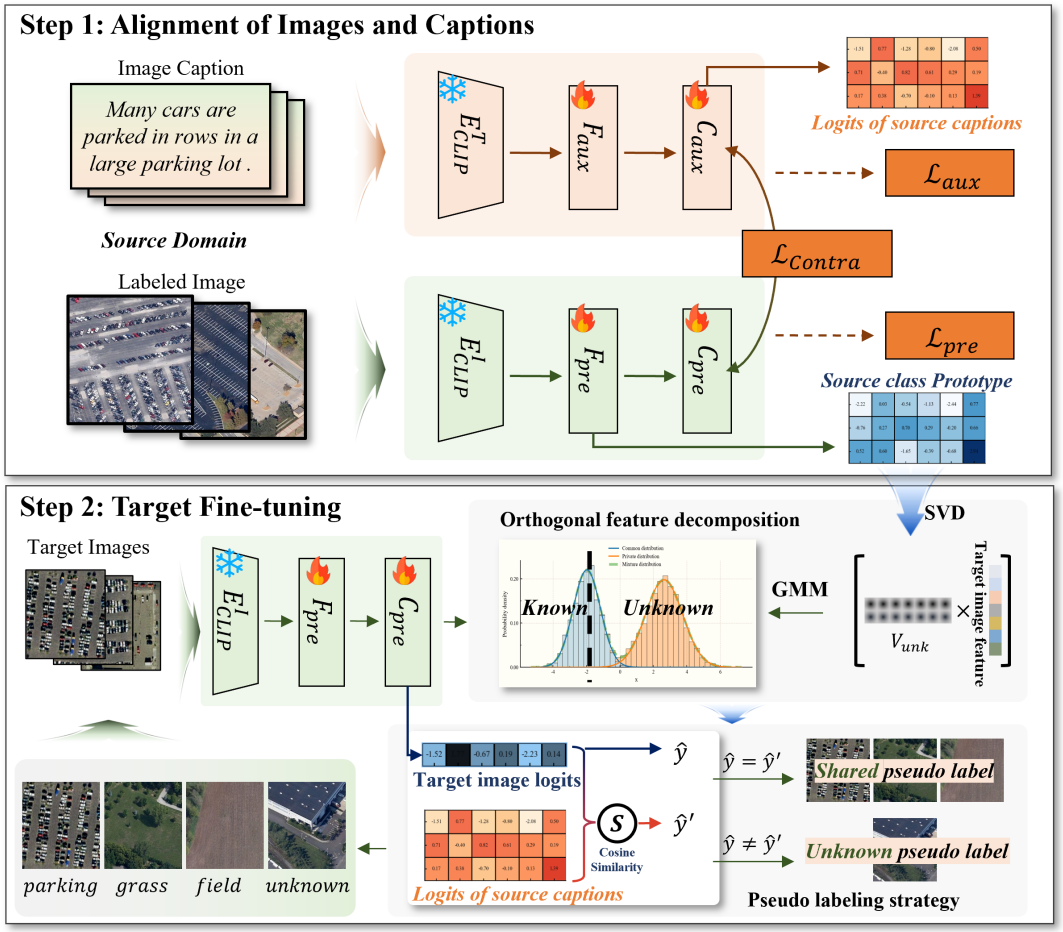


Fig. 2. Overall framework of the multisource unsupervised DA approach (MUniDA).

pseudolabeling strategy is designed to assign common labels to common data and unknown label to private data. Conveniently, the source domain comprises n_s labeled samples drawn from the distribution $P(X_{\text{cap}}, X_s, Y_s)$, where X_s denotes the images and X_{cap} denotes their captions. Similarly, the target domain X_t includes n_t images with a label space Y_t. The common label set $Y = Y_s \cap Y_t$ represents the shared classes. It is important to note that Y_t and Y are unknown in the UniDA setting, making these definitions purely notational for convenience. Meanwhile, we define the text encoder of CLIP as E^T_{CLIP} and the image encoder of CLIP as E^I_{CLIP}. For the two classifiers, we define the auxiliary classifier trained with textual data as C_{aux} and the primary classifier trained with images as C_{pri}. While defining the linear layers in the classifier without the final output layer as F_{aux} and F_{pri}, respectively. In the following, we will use the symbols that are defined as above.

A. Alignment of Images and Captions

To train the auxiliary classifier and the primary classifier, the source domain is adopted. Given an source image $x^i_s \in X_s$ and its caption x^i_{cap}, the x^i_s and x^i_{cap} can be represented as

$$\begin{aligned} f^i_s &= F_{\text{pri}}\left(E^I_{\text{CLIP}}\left(x^i_s\right)\right) \\ f^i_{\text{cap}} &= F_{\text{aux}}\left(E^T_{\text{CLIP}}\left(x^i_{\text{cap}}\right)\right) \end{aligned} \tag{1}$$

where f^i_s and f^i_{cap} represent the features of x^i_s and x^i_{cap}, respectively. They are obtained based on the transformation of the features from CLIP with F_{pri} and F_{aux}. Then the classification outputs of the x^i_s and x^i_{cap} can be obtained as

$$\begin{aligned} p^i_s &= C_{\text{pri}}\left(f^i_s\right) \\ p^i_{\text{cap}} &= C_{\text{aux}}\left(f^i_{\text{cap}}\right) \end{aligned} \tag{2}$$

where p^i_s and p^i_{cap} are logits that belong to $\mathbb{R}^N$, and N is equal to $|Y_s+1|$, which represents the number of classification space.

To align images and captions, we employ the contrastive learning framework from CLIP. This approach trains the primary and auxiliary classifier simultaneously by applying contrastive learning to their respective logits. The contrastive loss can be represented as

$$\mathcal{L}_{\text{Contra}} = -\sum_{i=1}^{b} \log \frac{\exp\left(p^i_s \cdot p^i_{\text{cap}}\right)/\tau}{\sum_{k=1}^{b} \mathbb{1}_{[k\neq i]} \exp\left(p^i_s \cdot p^k_{\text{cap}}\right)/\tau} \tag{3}$$

where b is batch size in training, and $\mathbb{1} \in \{0, 1\}$ represents an indicator function that equals 1 when $k \neq i$. τ is the temperature hyper-parameter that controls the scale of the contrastive loss. Without loss of generality, the classifiers are also trained using cross-entropy loss. The overall loss for aligning images and captions is represented as

$$\mathcal{L}_{\text{align}} = \mathcal{L}_{\text{pri}} + \mathcal{L}_{\text{aux}} + \lambda \mathcal{L}_{\text{Contra}} \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

where $\mathcal{L}_{\text{pre}}$ and $\mathcal{L}_{\text{aux}}$ are cross entropy losses for images and image captions, respectively. λ is a tradeoff parameter.

B. Target Fine-Tuning

The target fine-tuning method is mainly developed based on pseudolabels. To obtain the confident pseudolabels, the orthogonal feature decomposition strategy and pseudolabeling strategy are proposed. In the following, each strategy will be introduced in detail.

1) Orthogonal Feature Decomposition: The primary challenge in UniDA is distinguishing between known and unknown data. To address this, source image prototypes are orthogonally decomposed into known and unknown spaces using singular value decomposition (SVD). This decomposition allows for the analysis of feature distributions in these distinct spaces.

To make the theoretical basis of the orthogonal feature decomposition easy to follow, we simply introduce the theoretical basis of the SVD [16]. For convenience, we define A as a real matrix $m \times n$ with $m \geq n$. Then A can be represented as

$$A = U\Sigma V^T \tag{5}$$

where $\Sigma \in \mathbb{R}^{m\times n}$ is a diagonal matrix, which contains the nonnegative and unique singular values $\sigma_i, i = 1, \ldots, n$ with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n > 0$. $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ are unitary matrices, and they contain the left- and right-singular vectors of A, respectively. The singular vectors in both U and V correspond to the singular values σ.

With the SVD, it can divide A into knowledge-relevant space and nonrelevant space. Following [36], it points out that although target data might have shifted in the feature space, target-private data from unknown categories are likely to exhibit a higher proportion of components from the orthogonal complement space spanned by source model weights. However, compared with source model weights, the source class prototypes may be much more direct to distinguish the categories. In this article, we propose to decompose the source class prototypes. For convenience, we define C as the number of the classes in Y_s. The source class prototype can be represented as

$$f_S^k = \sum_{x_s^i \in X_s, y_s^i \in Y_s} \mathbb{1}\left(y_s^i, k\right) f_s^i / \sum_{y_s^i \in Y_s} \mathbb{1}\left(y_s^i, k\right), \quad k = 1, 2, \ldots, C \tag{6}$$

where $\mathbb{1} \in \{0, 1\}$ represents an indicator function that equals 1 when y_s^i is equal to k. Suppose $f_S^k \in \mathbb{R}^D$, where D is the number of dimensions. Then the source class prototypes can be decomposed with SVD as

$$\left[f_S^1, f_S^2, \ldots, f_S^C\right]^T = U\Sigma V^T \tag{7}$$

where $U \in \mathbb{R}^{C\times C}$, $\sum \in \mathbb{R}^{C\times D}$ and $V \in \mathbb{R}^{D\times D}$. According to the theoretical basis of SVD, the first C columns of V are relevant to the C number of known classes, and we can regard the remaining $D-C$ columns as the knowledge that are related to unknown classes due to the orthogonal characteristics. Thus, the known and unknown space can be represented as

$$V_{\text{knw}} = \text{span}\left\{\mathbf{v}_n|_{n=1,\ldots,C}\right\} \tag{8}$$

$$V_{\text{unk}} = \text{span}\left\{\mathbf{v}_n|_{n=C+1,\ldots,D}\right\}. \tag{9}$$

Then we can project the target sample to the known space and unknown space, respectively.

Given a target sample $x_t^i \in X_t$, its feature can be represented as $f_t^i = F_{\text{pre}}(E_{\text{CLIP}}^I(x_t^i))$. To maintain a consistent scale, the features are normalized using $L2$ normalization, expressed as $f_t^i = f_t^i / \|f_t^i\|_2$. Subsequently, the normalized target features are projected onto the orthogonal bases derived from SVD

$$\begin{aligned} \mathbf{F}_{\text{known}}^i &= V_{\text{knw}}^T\left(f_t^i\right) \\ \mathbf{F}_{\text{unknown}}^i &= V_{\text{unk}}^T\left(f_t^i\right). \end{aligned} \tag{10}$$

Following LEAD [36], we can determine whether the data belongs to common data or the private data by analyzing the distribution of the squared norm of unknown component vectors $\|\mathbf{F}_{\text{unknown}}\|_2$, and the distribution of $\|\mathbf{F}_{\text{unknown}}\|_2$ can be characterized by two prominent peaks that generally signify common and private data. Then a Gaussian mixture model (GMM) is adopted to represent the distribution of $\|\mathbf{F}_{\text{unknown}}\|_2$, which can be represented as $\|\mathbf{F}_{\text{unknown}}\|_2$

$$p\left(\|\mathbf{F}_{\text{unknown}}\|_2^2\right) = \pi_{\text{com}}\mathcal{N}\left(\mu_{\text{com}}, \sigma_{\text{com}}^2\right) + \pi_{\text{pri}}\mathcal{N}\left(\mu_{\text{pri}}, \sigma_{\text{pri}}^2\right) \tag{11}$$

where π_{com} and π_{pri} are mixture weights, satisfying $\pi_{\text{com}} + \pi_{\text{pri}} = 1$. By optimizing (11) with the Expectation–Maximization algorithm, we can obtain the two prominent peaks μ_{com} and μ_{pri}, which can denote the mean values for common and private data, respectively. From the empirical analysis in [36], μ_{com} is lower than μ_{pri}. Therefore, we assume that the image x_t^i has high confidence that it may belong to known classes if it $\|\mathbf{F}_{\text{unknown}}^i\|_2$ is smaller than μ_{com}.

2) Pseudolabeling Strategy: The absence of labels for target domain images necessitates the adoption of pseudolabeling to facilitate model fine-tuning on the target domain. This approach leverages high-confidence predictions as pseudolabels, enabling iterative improvements in model performance. The pseudolabeling procedure is outlined as follows.

Initially, the pretrained image model generates logits for the target domain images, which can be represented as

$$p_t^i = C_{\text{pri}}\left(f_t^i\right). \tag{12}$$

Subsequently, the softmax function is applied to the logits p_t^i to compute the probability distribution over all categories. The category with the highest probability is identified as *haty*, represented as

$$\hat{y} = \text{argmax}\left(\text{softmax}\left(p_t^i\right)\right). \tag{13}$$

To enhance the quality of pseudolabels for the common data, we propose a double-check mechanism for each target sample. This approach extends beyond relying solely on the predictions of the image encoder by integrating guidance from source domain image captions. These captions provide valuable semantic information that remains relevant for interpreting the target domain due to the shared visual characteristics between the domains. Specifically, we calculate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

the average logits of image captions for each class, denoted as $[p^1_{\text{Cap}}, p^2_{\text{Cap}}, \ldots, p^C_{\text{Cap}}]$, which can be represented as

$$p^k_{\text{Cap}} = \sum_{x^i_t \in X_t, y^i_t \in Y_t} \mathbb{1}\left(y^i_t, k\right) p^i_{\text{cap}} / \sum_{y^i_t \in Y_t} \mathbb{1}\left(y^i_t, k\right), \quad k = 1, 2, \ldots, C. \tag{14}$$

The logits of the ith image in class k are denoted as $p^{k,i}_t$. Intuitively, an image exhibits greater proximity to the image captions from its corresponding class [37]. By computing similarity scores between the logits of target image and that of the image captions, an auxiliary pseudolabel $\hat{y}'$ is derived by

$$\begin{aligned} \text{sim} &= p^i_t \cdot \left[p^1_{\text{Cap}}, p^2_{\text{Cap}}, \ldots, p^C_{\text{Cap}}\right] \\ \hat{y}' &= \text{argmax}\ (\text{softmax}\,(\text{sim})). \end{aligned} \tag{15}$$

The consistency between the prediction of the primary classifier and the auxiliary pseudolabel $\hat{y}'$ serves as a critical manner to assign confident labels to the common data. Specifically, if the prediction $\hat{y}$ from the primary classifier aligns with the $\hat{y}'$ derived from similarity, and the squared norm of the unknown component vector $\|\mathbf{F}_{\text{unknown}}\|_2$ is below the threshold μ_{com}, the image is confidently assigned to the class represented by $\hat{y}$ as its pseudolabel. To obtain the discriminative unknown samples, we select them not only from the private data but also regard the samples that cannot be predicted with consistent predictions between the prediction of the primary classifier and the auxiliary pseudolabel in common data as the candidates, since these samples may be distributed along the boundary between the samples with known class and those with unknown class. To ensure the discriminativess of private data, it is important to consider that the similarity between the prediction logits of the primary classifier and the logits of captions for a sample from the common data may be significantly higher than that for a sample from the private data. Consequently, samples lacking consistent predictions are assigned the unknown label by ranking their maximum similarity between the prediction logits of the primary classifier and the logits of captions in descending order. For balanced labeling, a fixed percentage, t, of these samples is selected for training, where t can be determined based on practical requirements. For convenience, we define the samples with pseudolabels, including the commons classes and unknown label, as $\hat{D}_t$, and the primary classifier can be fine-tuned with the following cross entropy loss:

$$\mathcal{L}_{\text{fine-tuning}} = -\sum_{x_t \in \hat{D}_t} \hat{y}\,(x_t) \log \hat{p}^i_t \tag{16}$$

where $\hat{y}(x_t)$ is the pseudolabel for $x_t \in \hat{D}_t$, $\hat{p}^i_t$ is the probability of x_t with the primary classifier.

IV. Experiments

This section provides a detailed overview of the experiments.

A. Datasets

In this article, four popular and benchmark remote sensing scene datasets are adopted, including the RSSCN7 [38], UC Merced [39], AID [40], and NWPU-RESISC45 [41] datasets. The RSSCN7 dataset includes seven scene categories, each containing 400 images of size 400 × 400 pixels, captured under varying seasonal and lighting conditions. The UC Merced dataset consists of 2100 images evenly distributed across 21 categories, with each image sized at 256×256 pixels and derived from aerial imagery with a 30-m spatial resolution. The AID dataset, a large-scale benchmark for aerial scene classification, includes 10 000 images across 30 diverse categories, with images sized at 600 × 600 pixels, varying in conditions and resolutions to ensure diversity. The NWPU-RESISC45 dataset comprises 31 500 images across 45 classes, with each class containing 700 images of 256 × 256 pixels, collected from various sensors and platforms under diverse conditions.



Fig. 3. Some images of five shared categories from four datasets. These images in the top row to the bottom row are from RSSCN7 dataset, UC Merced dataset, AID dataset, and NWPU-RESISC45 dataset, respectively.

Table I presents three UniDA tasks for remote sensing image scene classification. The RSSCN7 dataset serves as the source domain, enabling three cross-domain scenarios: RSSCN7->UC Merced, RSSCN7->AID, and RSSCN7->NWPU-RESISC45. To represent the relationship between the label space of target domain and that of the source domain, the Jaccard index is adopted to quantify the overlap between label sets of the source and target domains, denoted as $\xi = |Y|/|Y_s \cup Y_t|$, where $|Y|$ is the number of common classes and $|Y_s \cup Y_t|$ is the number of all the classes in both the source domain and the target domain. ξ, a rational number between 0 and 1, indicates the difficulty of the UniDA task; $\xi = 1$ corresponds to closed-set DA, while smaller values make the task more challenging.

Given the labels in source and target domains may not align precisely in definition, we adopt the strategy outlined in UniDA-MA [11] to define shared and private classes based on both semantic identity and visual similarity: classes with identical names are treated as shared, while others are categorized based on cross-domain visual affinity (see Fig. 3 and Table II). To quantify interclass correspondences, we compute pairwise cosine similarities between class prototypes, derived by averaging the [CLS] tokens extracted from the CLIP

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

TABLE I

THREE TASKS FOR REMOTE SENSING SCENE CLASSIFICATION WITH RSSCN7, UC MERCED, AID, AND NWPU-RESISC45

RSSCN->UC Merced	Total label sets	Shared label sets	Private label sets	ξ
source domain:RSSCN7 dataset target domain:UC Merced dataset	7 21	5 5	2 16	0.18
RSSCN->AID	Total label sets	Shared label sets	Private label sets	ξ
source domain:RSSCN7 dataset target domain:AID dataset	7 30	6 6	1 24	0.16
RSSCN->NWPU-RESISC45	Total label sets	Shared label sets	Private label sets	ξ
source domain:RSSCN7 dataset target domain:NWPU-RESISC45 dataset	7 45	6 6	1 39	0.12

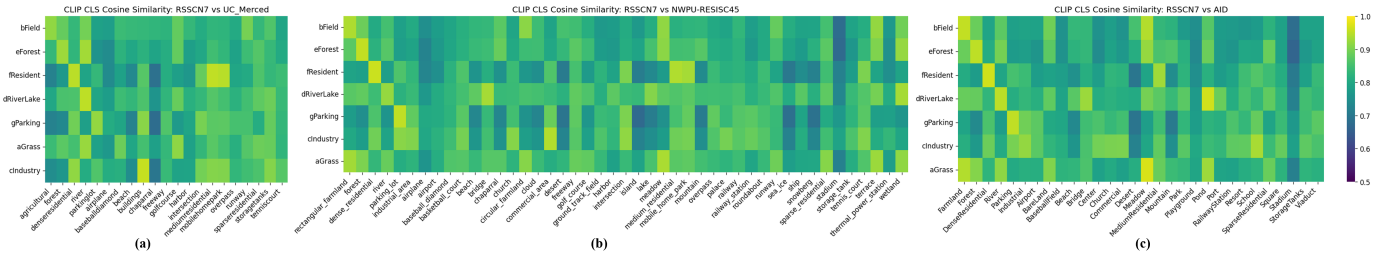


Fig. 4. Visualization of similarity metrics between shared and private classes of RSSCN7 and other datasets. (a) Similarity measurement diagram between categories of RSSCN7 and UC Merced dataset. (b) Similarity measurement diagram between categories of RSSCN7 and NWPU RESISC45 dataset. (c) Similarity measurement diagram between categories of RSSCN7 and AID dataset.

TABLE II

DEFINITION OF SHARED AND PRIVATE CLASSES IN THREE DA TASKS

Datasets	Shared class in source domain	Shared class in target domain
RSSCN7->UC Merced	Field, Forest, Resident, Riverlake, Parking	Agricultural, Forest, Dense residential, River, Parking lot
RSSCN7->AID	Field, Forest, Resident, Riverlake, Parking, Industry	Farmland, Forest, Dense residential, River, Parking, Industrial
RSSCN7->NWPU-RESISC45	Field, Forest, Resident, Riverlake, Parking, Industry	Rectangular farmland, Forest, Dense residential, River, Parking lot, Industrial area



(a) (b) (c)

Fig. 5. (a) Image: a345, Grass, RSSCN7 dataset. A long road divides green grass fields on both sides. (b) Image: f275, Resident, RSSCN7 dataset. Many houses lined up in a residential neighborhood. (c) Image: g004, Parking, RSSCN7 dataset. Many cars are parked in a large parking lot with different sections.

image encoder for each category. The resulting similarity maps (Fig. 4) reveal that single source classes frequently exhibit high similarity to multiple, finer-grained target classes along the main diagonal. These many-to-many semantic relationships reflect the multilevel granularity variations inherent in remote sensing data, which significantly complicate domain-adaptive feature disentanglement.

We use gpt-4o to generate image captions. To guarantee that the caption can provide necessary information, we provide a template prompt for generation. i.e., "This image is about the <class> area. Please use plain English to generate an image caption without phrases that like "An aerial view of" and "The image shows." While "<class>" or "<class> area" must be contained in the caption. Please only focus on <class> related information." In the query message, "<class>" denote a certain class like farmland, river and so on. "Without phrases like "An aerial view of," "The image shows," and so on" is used to avoid redundant words which may appear in all the classes and dilute distinct information. "<class>" or "<class> area" must be contained in the caption" is used to guarantee that its corresponding label is included in the sentence. Some examples of captions are shown in Fig. 5.

B. Comparison Algorithms

To evaluate the efficacy of our approach, the following methods are used for comparison.

1) *I-UAN [42]:* It introduces a novel approach using pseudomargin vectors to efficiently identify the common label set between domains by estimating each source class's probability of belonging to it. It incorporates pseudomargin vectors and target pseudomargin register for better domain alignment, enhancing UniDA perfor-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

mance through class-wise and sample-wise weighting mechanisms.

2) *UDA [10]:* It first introduces a framework that addresses the challenge of adapting models from source domain to target domain with unknown label sets. The framework is designed to handle varying label spaces and identify target samples belonging to categories not present in the source domain. It helps align known classes and reject unknown ones.
3) *CMU [43]:* It proposes a novel approach where transferable weights are estimated using a combination of complementary uncertainty measures: entropy, confidence, and consistency. The transferability measure accurately quantifies the inclination of a target example to the open classes.
4) *UniDA-MA [11]:* It is the first time that the setting of UniDA has been used for remote sensing image scene classification. A transferable weight is used to distinguish the shared and private label sets in each domain, helping promote the adaptation in the automatically discovered shared label set and recognize the "unknown" samples successfully.
5) *LEAD [36]:* This work proposes a new idea of learning decomposition, which decouples features into source-known and -unknown components to identify target private data. Technically, LEAD leverages the orthogonal decomposition analysis for feature decomposition. Then, it builds instance-level decision boundaries to adaptively identify target-private data.
6) *MLNet [13]:* In this method, confidence-guided invariant feature learning with self-adaptive neighbor selection is designed to reduce the intradomain variations for more generalizable feature representation. By using the cross-domain mixup scheme for better unknown-class identification, the proposed method compensates for the misidentified known class errors by mutual learning between the closed-set and open-set classifiers.

C. Implementation Details

The experiments were conducted on a single RTX-4090 GPU using PyTorch, with a batch size of 36. The CLIP pretrained model, employing the VIT-B/16 image encoder [44], was utilized for the implementation. During the pretraining phase, the AdamW [45] optimizer was applied with an initial learning rate of 2×10^{-6}, training the model for 600 iterations. The learning rate was reduced by a factor of 0.1 after 200 iterations. The hyperparameters λ and τ were set to 0.05 and 1, respectively. For target fine-tuning, the AdamW optimizer was used with a learning rate of 5×10^{-7}. Pseudolabels were alternately updated, and the model was fine-tuned for a total of eight epochs.

To evaluate the classification performance comprehensively, we employ the H-score metric, which integrates accuracy assessments for both common classes (a_Y) and private classes ($a_{\overline{Y}_t}$). The H-score is computed with H-score $= 2 \times (a_Y \times a_{\overline{Y}_t})/(a_Y + a_{\overline{Y}_t})$.

D. Experimental Results

1) Results on RSSCN7 → UC Merced: The RSSCN7 → UC Merced adaptation task demonstrates the superior performance of the proposed approach, as detailed in Table III. The method achieves notable advancements across all evaluated metrics, surpassing the leading baseline, LEAD, by an H-score margin of 8.72%. This improvement highlights the method's effectiveness in unknown class accuracy, shared class accuracy, average accuracy, and overall H-score. The results underscore the efficacy of the comprehensive pseudolabel selection strategy, which integrates image classifier predictions, cross-modality knowledge, and orthogonal feature decomposition. This approach proves to be more effective than the feature decomposition employed in LEAD, the self-adaptive neighbor selection in MLNet, the sample-level transferable weight in UniDA-MA, and other strategies such as entropy-based methods in UDA and pseudomargin vectors in I-UAN for remote sensing scene classification.

2) Results on RSSCN7 → AID: The RSSCN7 → AID adaptation task results, summarized in Table IV, demonstrate the proposed method's superior performance. It achieves an average accuracy of 83.00% and an H-score of 77.94%, outperforming existing methods by an H-score margin of 11.44%. The method's balanced performance in detecting shared and unknown classes highlights its robustness and practical utility in addressing the complexities of remote sensing image scene classification.

3) Results on RSSCN7 → NWPU: The RSSCN7 → NWPU task results, presented in Table V, demonstrate the superior performance of the proposed approach, achieving an H-score of 68.04%. This performance surpasses all comparative methods, with notable improvements in distinguishing challenging categories. Specifically, the method achieves H-score improvements of 23.28% and 5.57% over LEAD and MLNet in the "River" category, respectively, and 49.43% and 7.29% in the "Industries" category. These findings underscore the method's robustness and effectiveness in addressing both label shift and category shift, highlighting its practical utility in complex remote sensing scene classification tasks.

Despite the variability of ξ across the three tasks, the proposed method consistently outperforms state-of-the-art approaches. These results validate the method's practical applicability in real-world scenarios, even when the overlap between source and target domains varies significantly.

E. Analysis of Computational Complexity

For the proposed method, there are two stages. One is pertaining to the source domain. The other one is to fine-tune the pretrained model. For the first step, we add two linear layers following the CLIP encoder for each classifier, i.e., image classifier and the text classifier, and the CLIP encoder is frozen. For convenience, we define the output dimension of CLIP for images as C_{Clip}, and define the batch size at each iteration as B. Meanwhile, we define the output of the hidden dimension as D_h and the computational complexity of CLIP as $\text{Cost}_{\text{CLIP}}$. The computational complexity of the first stage can be represented as $O(B(\text{Cost}_{\text{CLIP}} + 2D_hC + 2C_{\text{Clip}}D_h))$ at each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

TABLE III
EXPERIMENTAL RESULTS COMPARISON OF DIFFERENT METHODS ON RSSCN7->UC MERCED

Methods	Category						Avg_shared	Avg	H-score
	Farmland	Forests	Dense Residential	Rivers	Parking	Unknown			
I-UAN	92.00	97.00	72.00	70.00	30.00	11.63	72.20	62.10	20.03
UDA	87.00	98.00	**83.00**	84.00	68.00	1.13	84.00	70.19	2.23
CMU	89.00	99.00	68.00	78.00	**100.00**	12.25	86.80	74.38	21.47
UniDA-MA	85.00	98.00	74.00	79.00	**100.00**	14.63	87.20	75.11	25.06
LEAD	85.00	**100.00**	46.00	68.00	98.00	45.88	79.40	73.81	58.15
MLNet	92.00	98.00	41.00	58.00	98.00	61.06	77.40	74.68	68.27
Ours	**97.00**	98.00	78.00	**86.00**	94.00	**66.94**	**90.60**	**86.66**	**76.99**

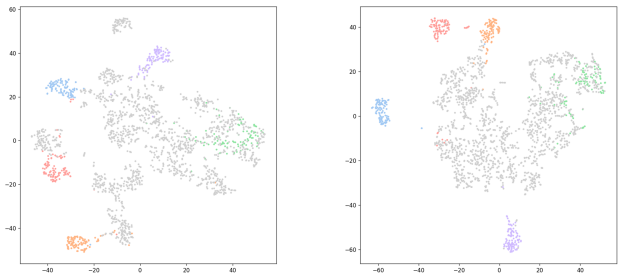
TABLE IV
EXPERIMENTAL RESULTS COMPARISON OF DIFFERENT METHODS ON RSSCN7->AID

Methods	Category							Avg_shared	Avg	H-score
	Farmland	Forests	Dense Residential	Rivers	Parking	Industries	Unknown			
I-UAN	93.24	99.60	94.63	59.27	81.79	61.79	10.84	81.72	71.60	19.14
UDA	92.16	99.60	**97.56**	47.56	93.33	68.21	2.52	83.07	71.56	4.89
CMU	83.51	98.00	89.51	**59.51**	91.79	80.77	10.51	83.85	73.37	18.68
UniDA-MA	92.70	**100.00**	94.63	58.78	94.36	70.51	13.48	**85.16**	74.92	23.28
LEAD	91.08	**100.00**	87.80	57.56	98.72	44.36	56.93	79.92	76.64	66.50
MLNet	79.19	96.80	90.98	27.07	92.82	65.13	52.38	75.33	72.05	61.79
Ours	**95.41**	98.00	97.32	30.00	**99.49**	**88.72**	**72.10**	84.82	**83.00**	**77.94**

TABLE V
EXPERIMENTAL RESULTS COMPARISON OF DIFFERENT METHODS ON RSSCN7->NWPU-RESISC45

Methods	Category							Avg_shared	Avg	H-score
	Farmland	Forests	Dense Residential	Rivers	Parking	Industries	Unknown			
I-UAN	85.71	98.14	96.43	35.57	28.43	**89.00**	5.52	72.21	62.69	10.26
UDA	77.29	97.29	88.14	40.86	90.86	76.71	2.93	78.52	67.72	5.65
CMU	80.00	95.57	72.86	**47.71**	84.14	80.43	12.18	76.79	67.56	21.03
UniDA-MA	84.57	94.29	88.43	40.00	92.00	80.14	14.36	**79.90**	70.54	24.34
LEAD	65.43	**98.29**	89.43	10.86	47.57	19.57	50.09	55.19	54.46	52.52
MLNet	65.71	86.86	59.00	28.57	54.14	61.71	**67.97**	59.33	60.57	63.36
Ours	**96.29**	79.86	**97.71**	34.14	**97.86**	69.00	59.67	79.14	**76.36**	**68.04**

iteration. While for the second step, the main computational cost is the classifier fine-tuning and the SVD to decompose the source class prototype. Hence, compared with the first stage, the additional computation is SVD, whose computational complexity is $O(D_h C^2)$. Thus, for one iteration of the proposed method, the computational complexity can be represented as $O(2B(\text{Cost}_{\text{CLIP}} + 2D_h C + 2C_{\text{Clip}} D_h) + D_h C^2)$. We should note that there are no expensive computation components in the proposed method. Although the computational cost of SVD is expensive, the number of C is usually a small number. Generally, the computational time and the computational complexity are consistent. Therefore the computational time is also fast at each iteration. For the memory, the architecture of the proposed method mainly contains CLIP and linear classifier layers. The additional parameters mainly are the linear classifier layers. If we define the number of the parameters of CLIP is M_{Clip}. The parameters of the proposed method can be represented as $M_{\text{Clip}} + 2(D_h C + 2C_{\text{Clip}} D_h)$.



(a) (b)

Fig. 6. t-SNE visualization of target domain image logits in RSSCN7 → UC Merced task. gray plots are "unknown" samples. (a) source only. (b) after adaptation.

F. Model Analysis

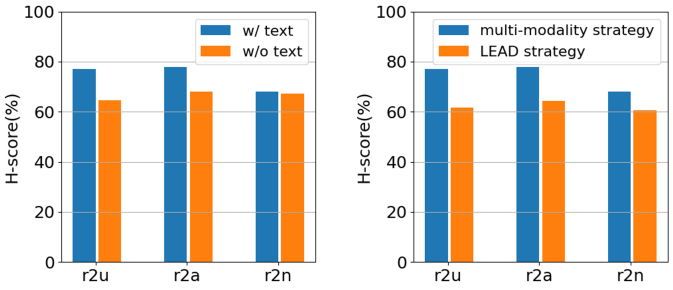
1) Feature Distribution Analysis: The t-SNE is used to visualize the target image logits distribution. As is shown in Fig. 6(a), target private samples are attached near the shared samples in RSSCN7 → UC Merced task. After DA [Fig. 6(b)], most target private samples are separated from the shared samples. This phenomenon demonstrates that our proposed MUniDA has superior ability to distinguish known samples from unknown samples.

2) Effect of Text Guidance: To evaluate the impact of text guidance, we conduct ablation studies by removing image captions during both training and inference phases. This results in performance drops of 12.53%, 9.92%, and 0.88% on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

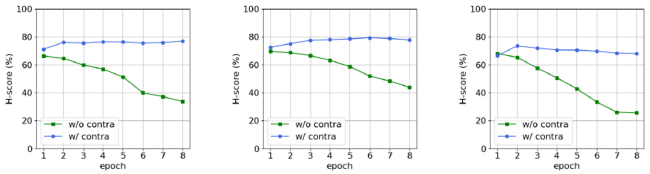
TABLE VI
EXPERIMENTAL RESULTS COMPARISON OF CAPTION NUMBER PER IMAGE ON THE THREE DATASETS

image-caption number	RSSCN7->UC Merced			RSSCN7->AID			RSSCN7->NWPU-RESISC45		
	Unknown	Avg_shared	H-score	Unknown	Avg_shared	H-score	Unknown	Avg_shared	H-score
1	66.94	**90.60**	**76.99**	72.10	**84.82**	77.94	59.67	79.14	68.04
3	62.75	88.20	73.33	**77.11**	80.75	**78.89**	60.23	80.12	68.76
5	**67.63**	87.00	76.10	75.23	81.12	78.06	**63.69**	**85.45**	**72.98**



(a) (b)

Fig. 7. Ablation studies. (a) Image captions. (b) Pseudolabel selecting strategy.



(a) (b) (c)

Fig. 8. Ablation study for contrastive learning. (a) Without contrastive learning on RSSCN7 → UC Merced. (b) Without contrastive learning on RSSCN7 → AID. (c) Without contrastive learning on RSSCN7 → NWPU.

RSSCN7 → UC Merced, RSSCN7 → AID and RSSCN7 → NWPU datasets, respectively, as is presented in Fig. 7(a). This performance degradation shows that invariant textual information can mitigate domain shift effects. Source image captions serve as a crucial bridge between source and target domains.

3) Pseudolabel Selecting Strategy: We propose a comprehensive pseudolabel selection strategy that integrates information from three key components including image classifier's predictions, similarity between images and descriptive texts, and feature decomposition. Fig. 7(b) demonstrates the superiority of our proposed method. Specifically, our multimodality strategy achieves 15.32%, 13.52%, and 7.33% H-score improvements over the LEAD method [36] on the RSSCN7 → UC Merced, RSSCN7 → AID, and RSSCN7 → NWPU datasets, respectively, validating its effectiveness in pseudolabel selection.

4) Caption Number Ablation: Existing image caption datasets often consist of image text pairs consisting of an image and five corresponding image captions. We therefore examine how this number affects performance. As shown in Table VI, experiments are conducted on 1, 3, and 5 captions, respectively. Because the improvement across the three benchmarks is limited and unstable and the computational burden scale linearly, we retain the single-caption configuration as the default balance between accuracy and efficiency.

5) Effect of Contrastive Learning: To assess the impact of contrastive learning between images and their corresponding captions, we conduct ablation studies by removing the contrastive learning module. It can be seen in Fig. 8 that the classification performance significantly degrades across all three datasets. It means that image captions cannot offer proper guidance to classification without being aligned with its corresponding images. Contrastive learning approach serves to align image-text pairs, enabling the model to better comprehend visual concepts through their textual contexts and thereby enhancing classification performance.

TABLE VII
H-SCORE(%) COMPARISON OF CONTRASTIVE LEARNING VARIATIONS

Methods		r2u	r2a	r2n
text	CL level			
image caption	logits	76.99	77.94	68.04
image caption	feature	50.31	31.54	28.34
prompt	logits	30.80	27.90	25.90
prompt	feature	29.69	28.02	26.58

6) Contrastive Learning Variations Analysis: We investigate two key factors in contrastive learning: the type of descriptive text (image captions versus prompts) and the layer for contrastive learning (feature layer versus logits layer). Image captions use a sentence with class name to depict corresponding images. "A photo of [CLASS]" is a prompt, where the [CLASS] token denotes class name of each category. In theory, we can use either image captions or prompts as descriptive texts to perform contrastive learning with images. And we can also perform contrastive learning on feature layer or logits layer. So, we investigate into which combination can achieve the best result. Table VII shows contrastive learning variations. Image captions consistently outperform prompts across all datasets. This performance gap arises because prompts share excessive common vocabulary across classes (e.g., "A photo of"), which may dilute discriminative information. For image caption, we observe that logits-based contrastive learning significantly outperforms feature-level contrastive learning, achieving H-score improvements of 26.68%, 46.40%, and 39.70% on the three datasets, respectively. Compared with the alignment at feature level, the decision level can more effectively and directly improve the generalization ability, since the alignment at feature level cannot guarantee that the unseen images are similar with the existing images while the decision level can prevent biased estimation of feature importance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

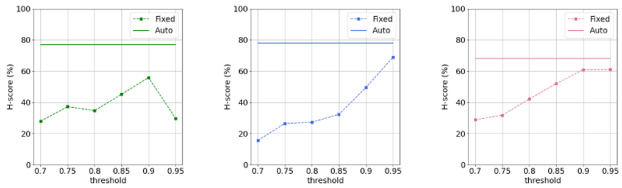


Fig. 9. Auto known threshold versus fixed known threshold. (a) RSSCN7 → UC Merced. (b) RSSCN7 → AID. (c) RSSCN7 → NWPU.

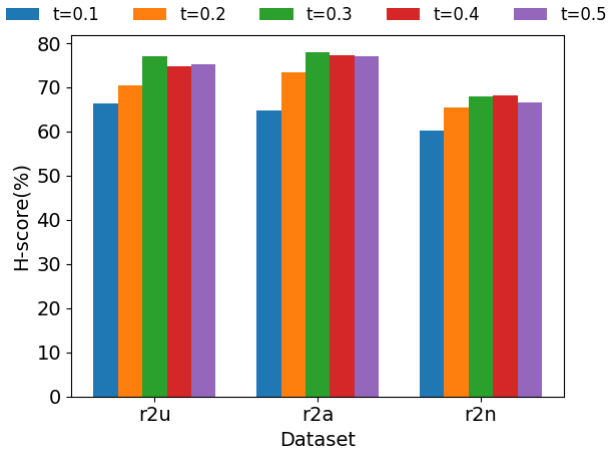


Fig. 10. Impact of t to the model performance on RSSCN7 → UC Merced, RSSCN7 → AID and RSSCN7 → NWPU datasets.

7) Known Decision Threshold Analysis: To evaluate the effectiveness of our feature decomposition-based thresholding approach, we compare it against conventional fixed thresholds ranging from 0.7 to 0.95 (in increments of 0.05). As shown in Fig. 9, classifier-based thresholds consistently underperform our decomposition strategy, with their best H-scores being 21.04%, 9.05%, and 7.02% lower on the RSSCN7 → UC Merced, RSSCN7 → AID, and RSSCN7 → NWPU datasets, respectively. This performance gap arises because high thresholds tend to over-classify samples as "unknown," while low thresholds increase false positives in known categories. Fixed thresholds demonstrate limited adaptability to category shift. Even though the value 0.9 is acceptable for all the three datasets, it is still inferior to our feature decomposition based auto threshold.

8) Unknown Decision Threshold Analysis: To explore impact of unknown decision threshold t, we evaluate its performance across values ranging from 0.1 to 0.5 (in increments of 0.1) using H-score as the evaluation metric. As shown in Fig. 10, $t = 0.3$ is the best choice for RSSCN7 → UC Merced ($\xi = 0.18$) and RSSCN7 → AID ($\xi = 0.16$) datasets, which outperform the next best values by 1.83% and 0.57% on H-score, respectively. For RSSCN7 → NWPU ($\xi = 0.12$), $t = 0.4$ yields the highest H-score. However, $t = 0.3$ remains competitive, trailing by only 0.17%. Further analysis reveals that the superior performance of $t = 0.4$ on NWPU in H-score primarily stems from enhanced unknown accuracy at the expense of certain known classes. Specifically, only 6.10% of "river" samples are correctly classified at this setting. This tradeoff demonstrates that $t = 0.3$ strikes an optimal balance across all three datasets, maintaining robust performance despite varying class overlaps between source and target domains. When prioritizing shared class accuracy, reducing t below 0.3 can be beneficial. Our proposed method outperforms all the compared methods on shared class accuracy, average accuracy and H-score when t is set to 0.2. The unknown decision threshold t is fixed and it picks out a fixed proportion of samples that have not yet been assigned pseudolabels. Because the known decision threshold is adaptive, the size of the unknown set automatically adjusts with changes in the known threshold, eliminating any need to make the unknown threshold adaptive as well.

V. Conclusion

We introduce MUniDA, a comprehensive prediction framework designed for UniDA in remote sensing image scene classification. This framework contains two important steps. The first step is the alignment of images and captions on the source domain based on CLIP. Meanwhile, an auxiliary classifier and a primary classifier following the CLIP are trained with captions and images, respectively. Then based on alignment, a target fine-tuning method is developed to adapt the target task by fine-tuning with pseudolabels, which are obtained with orthogonal feature decomposition and a pseudolabeling strategy. The orthogonal feature decomposition is adopted with SVD on source class prototype to distinguish the target data into common data and private data. While the pseudolabeling strategy attempts to select the samples with confident common classes from common data and select the samples that may belong to the private classes by measuring the uncertainty and the similarity between the logits of the primary classifier and the logits of the captions from the auxiliary classifier. The extensive results show that the proposed method outperforms the state-of-the-art methods significantly, which effectively demonstrate the promising ability of muldality data for remote sensing scene classification in UniDA scenario.

References

[1] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[2] X. Yao, J. Han, G. Cheng, X. Qian, and L. Guo, "Semantic annotation of high-resolution satellite images via weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3660–3671, Jun. 2016.

[3] J. Xie, N. He, L. Fang, and A. Plaza, "Scale-free convolutional neural network for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6916–6928, Sep. 2019.

[4] Y. Yu, X. Li, and F. Liu, "Attention GANs: Unsupervised deep feature learning for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 519–531, Jan. 2020.

[5] K. Saenko, B. Kulis, M. Fritz, and T. Darrell, "Adapting visual category models to new domains," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2010, pp. 213–226.

[6] J. Guo, J. Yang, H. Yue, X. Liu, and K. Li, "Unsupervised domain-invariant feature learning for cloud detection of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5405715, doi: 10.1109/TGRS.2021.3120001.

[7] Q. Xu, C. Ouyang, T. Jiang, X. Yuan, X. Fan, and D. Cheng, "MFFENet and ADANet: A robust deep transfer learning method and its application in high precision and fast cross-scene recognition of earthquake-induced landslides," *Landslides*, vol. 19, no. 7, pp. 1617–1647, Jul. 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.

[8] Z. Cao, L. Ma, M. Long, and J. Wang, "Partial adversarial domain adaptation," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 135–150.

[9] K. Saito, S. Yamamoto, Y. Ushiku, and T. Harada, "Open set domain adaptation by backpropagation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 153–168.

[10] K. You, M. Long, Z. Cao, J. Wang, and M. I. Jordan, "Universal domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 2720–2729.

[11] Q. Xu, Y. Shi, X. Yuan, and X. X. Zhu, "Universal domain adaptation for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4700515, doi: 10.1109/TGRS.2023.3235988.

[12] J. Guo et al., "C^3DA: A universal domain adaptation method for scene classification from remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[13] Y. Lu, M. Shen, J. Andy, X. Xie, and J. Lai, "MLNet: Mutual learning network with neighborhood invariance for universal domain adaptation," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, 2024, pp. 3900–3908.

[14] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[15] E. Othman, Y. Bazi, F. Melgani, H. Alhichri, N. Alajlan, and M. Zuair, "Domain adaptation network for cross-scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4441–4456, Aug. 2017.

[16] S. Weiss, I. K. Proudler, G. Barbarino, J. Pestana, and J. G. McWhirter, "On properties and structure of the analytic singular value decomposition," *IEEE Trans. Signal Process.*, vol. 72, pp. 2260–2275, 2024.

[17] Q. Xu, X. Yuan, and C. Ouyang, "Class-aware domain adaptation for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4500317, doi: 10.1109/TGRS.2020.3031926.

[18] D. Wittich and F. Rottensteiner, "Appearance based deep domain adaptation for the classification of aerial images," 2021, *arXiv:2108.07779*.

[19] S. Lyu et al., "Deep learning based domain adaptation methods in remote sensing: A comprehensive survey," 2025, *arXiv:2510.15615*.

[20] J. Zeng et al., "Unsupervised domain adaptation for remote sensing semantic segmentation with the 2D discrete wavelet transform," *Sci. Rep.*, vol. 14, no. 1, p. 23552, Oct. 2024.

[21] X. Ma, X. Zhang, X. Ding, M.-O. Pun, and S. Ma, "Decomposition-based unsupervised domain adaptation for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5645118, doi: 10.1109/TGRS.2024.3483283.

[22] Y. Huang, J. Peng, G. Zhang, W. Sun, N. Chen, and Q. Du, "Adversarial domain adaptation network with calibrated prototype and dynamic instance convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514613, doi: 10.1109/TGRS.2024.3387990.

[23] T. Yang, S. Xiao, J. Qu, W. Dong, Q. Du, and Y. Li, "Graph embedding interclass relation-aware adaptive network for cross-scene classification of multisource remote sensing data," *IEEE Trans. Image Process.*, vol. 33, pp. 4459–4474, 2024.

[24] K. Shi, W. Wang, Q. Liu, and L. Xiao, "Topological information aggregation network for few-shot cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501015, doi: 10.1109/TGRS.2024.3516772.

[25] Z. Cao, K. You, M. Long, J. Wang, and Q. Yang, "Learning to transfer examples for partial domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2980–2989.

[26] J. Liang, Y. Wang, D. Hu, R. He, and J. Feng, "A balanced and uncertainty-aware approach for partial domain adaptation," Tech. Rep., 2020.

[27] J. Leng, Z. Chen, H. Mu, T. Liu, H. Chen, and G. Wang, "PPLM-Net: Partial patch local masking net for remote sensing image unsupervised domain adaptation classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17021–17035, 2024.

[28] Y. Zhu, J. Ai, L. Wu, D. Guo, W. Jia, and R. Hong, "An active multi-target domain adaptation strategy: Progressive class prototype rectification," *IEEE Trans. Multimedia*, vol. 27, pp. 1874–1886, 2025.

[29] J. Chen and X. Wang, "Open set few-shot remote sensing scene classification based on a multiorder graph convolutional network and domain adaptation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4709517, doi: 10.1109/TGRS.2022.3222449.

[30] Y. Guo, B. Yang, L. Du, and J. Chen, "Open-set domain adaptation via adaptive separation and two-level alignment with application in SAR target recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 25276–25290, 2025.

[31] A. Wang, R. Gomes, and P. F. Rozario, "OSDA-ST: Advancing open set domain adaptation through selective thresholding in remote sensing," in *Proc. IEEE Int. Conf. Big Data (BigData)*, Dec. 2024, pp. 7563–7570.

[32] Q. Li, Y. Wen, J. Zheng, Y. Zhang, and H. Fu, "HyUniDA: Breaking label set constraints for universal domain adaptation in cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5518415, doi: 10.1109/TGRS.2024.3400959.

[33] Q. Li, Y. Zhang, J. Zheng, Y. Zhang, J. Huang, and H. Fu, "Boosting universal domain adaptation in remote sensing with dual-classifiers consistency discrimination and cross-domain feature mixup," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5515015, doi: 10.1109/TGRS.2025.3571747.

[34] Q. Wu, Y. Liu, Y. Zhong, K. Lin, X. Ma, and M.-O. Pun, "Source-free multitarget unsupervised domain adaptation for cross-city local climate zone classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5407521, doi: 10.1109/TGRS.2025.3615979.

[35] Bi. Wang, F. Deng, Z. Chen, Z. Yu, and Y. Liu, "Prototype-based pseudo-label denoising for source-free domain adaptation in remote sensing semantic segmentation," 2025, *arXiv:2509.16942*.

[36] S. Qu et al., "LEAD: Learning decomposition for source-free universal domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 23334–23343.

[37] R. Kang, Y. Song, G. Gkioxari, and P. Perona, "Is clip ideal? No. Can we fix it? Yes!," 2025, *arXiv:2503.08723*.

[38] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.

[39] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2010, pp. 270–279.

[40] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[41] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[42] Y. Yin, Z. Yang, X. Wu, and H. Hu, "Pseudo-margin-based universal domain adaptation," *Knowl.-Based Syst.*, vol. 229, Oct. 2021, Art. no. 107315.

[43] B. Fu, Z. Cao, M. Long, and J. Wang, "Learning to detect open classes for universal domain adaptation," in *Proc. ECCV*, Aug. 2020, pp. 567–583.

[44] D. Alexey, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv: 2010.11929*.

[45] I. Loshchilov, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:33:02 UTC from IEEE Xplore. Restrictions apply.